

# **CS4023D Artificial Intelligence**

## **Assignment – 1**

**Group Name : MindMesh**

**3-SAT problem**

**Group Members :**

|                          |           |
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## Problem Statement

**3-SAT (3-Satisfiability)** is a well-known problem in computational complexity theory and is one of the classic examples of an **NP-Complete** problem. It belongs to the broader family of **SAT (Satisfiability)** problems.

SAT stands for **Boolean Satisfiability**. Given a Boolean formula expressed in **Conjunctive Normal Form (CNF)**—that is, a conjunction (AND) of clauses, where each clause is a OR of literals—the goal is to determine whether there exists an assignment of truth values (true/false) to the literals such that the entire formula evaluates to **true**.

**3-SAT** is a special case where each clause has 3 literals and is in CNF. The problem statement is as follows:

**“Given a Boolean formula in 3-CNF form, determine whether there exists an assignment of truth values to the literals so that the formula evaluates to true.”**

## State Space Representation

The state space of the problem is represented as a 4-tuple, (N, A, S, GD) which is defined as follows:

- **N (Set of States)**

We assume that our Boolean formula has  $n$  distinct literals which are

$x_1, x_2, x_3, \dots, x_n$  and since each clause has 3 variables there will be  $k$  clauses.

$$N = \{ (x_1, x_2, \dots, x_n) \mid \forall i \in [1, n], x_i \in \{0, 1, -\} \}$$

Where:

**0**  $\rightarrow$  Literal assigned **false**

**1**  $\rightarrow$  Literal assigned **true**

**–**  $\rightarrow$  Literal is **unassigned**

- **A (Set of Actions)**

There are 2 possible actions from any given state. Choose a literal  $x_i$  (un-assigned) and assign it the truth value **true(1)** or **false(0)**.

At level 1 we assign 1 or 0 to  $x_1$  similarly at level 2 we assign 1 or 0 to  $x_2$  like that we proceed.

Assign true

We are at level  $i$ ,  $(x_1, x_2, x_3, \dots, x_i, \dots, x_n) \rightarrow (x_1, x_2, x_3, \dots, 1, \dots, x_n)$

Assign false

We are at level  $i$ ,  $(x_1, x_2, x_3, \dots, x_i, \dots, x_n) \rightarrow (x_1, x_2, x_3, \dots, 0, \dots, x_n)$

- **S (Set of Start State(s))**

Initially all the literals in the Boolean formula will be un-assigned.

$S = \{(x_1, x_2, \dots, x_n) : \forall i \in [1, n], x_i = -\}$

- **GD (Set of Goal State(s))**

We say that a formula is satisfiable if it has is true, meaning all the clauses are true.

$GD = \{(x_1, x_2, \dots, x_n) \mid \forall x_i \in \{0, 1\}, \text{ and the assignment satisfies all clauses } c_j \in C\}$

Here,  $C$  is the set of clauses in the formula, and each  $c_j$  must evaluate to true under the assignment  $(x_1, x_2, \dots, x_n)$ .

## Example Instance of State Space

Let the Boolean formula in 3-CNF be:  $f = (x_1 \vee x_2 \vee x_3) \wedge (x_1 \vee x_2 \vee \neg x_3)$

Initial State =  $(-, -, -)$  meaning  $x_1 = -$ ,  $x_2 = -$  and  $x_3 = -$ .

Goal State =  $(x_1, x_2, x_3)$  such that the 2 clauses are true.

The state space diagram for the above problem is:

